AI-POWERED RESUME RANKER

Internship Project Report

1. Introduction

This project implements an AI-assisted resume ranking prototype. It accepts a job description and a PDF resume, extracts the resume text, applies NLP preprocessing, and calculates a similarity score. The project is based on the internship brief, which specifies PDF text extraction, SpaCy/Sklearn-style NLP processing, TF-IDF vectorization, keyword-based scoring, ranking, and a web interface.

2. Abstract

The system uses TF-IDF to represent the resume and job description as numerical vectors. Cosine similarity is then used to estimate how closely the resume content matches the job requirements. The web interface displays the percentage score along with matched and missing terms. This approach is lightweight, explainable, requires no API key, and can be extended with semantic embeddings or transformer models.

3. Tools Used

Python, Flask, pypdf, scikit-learn, HTML, CSS, TF-IDF and cosine similarity. Development can be performed in VS Code or any Python IDE. The application runs locally through Flask.

4. Steps Involved in Building the Project

1) Create a Flask web application. 2) Accept a PDF resume and job description. 3) Extract text from the PDF. 4) Clean the text and remove common stop words. 5) Convert both texts into TF-IDF vectors using unigram and bigram features. 6) Calculate cosine similarity and convert it to a percentage. 7) Identify overlapping and missing terms for explainability. 8) Display the score and keyword information through the browser UI. 9) Add unit tests, documentation and a reproducible requirements file.

5. Conclusion

The completed prototype demonstrates a practical NLP workflow for resume-job matching. It is intentionally transparent so that the score can be explained during an interview. It should be treated as decision-support software rather than an automatic hiring decision. Future improvements could include sentence embeddings, skill taxonomies, experience weighting, multilingual processing, authentication, database storage and deployment.

Deliverable	Included
Source code	app.py, utils.py
Web UI	Flask template + CSS
Documentation	README.md
Testing	tests/test_utils.py
Report	REPORT.pdf